Longitudinal patterns of antiretroviral treatment interruptions in a high burden, low-resource setting in South Africa: a retrospective cohort study

Claire M Keene, Jonathan Euvrard, Tali Cassidy, Mike English, Jacob McKnight, Ingrid T Katz, Catherine Orrell

Summary

Background Non-retention undermines the potential of antiretroviral therapy (ART) to improve individual and public health HIV outcomes in South Africa, where routine ART care is largely provided through primary care clinics or differentiated service delivery models (such as adherence clubs). We aimed to describe long-term patterns of treatment interruption and re-engagement in a large routine cohort.

Methods In this retrospective cohort study, we used demographic, mortality, dispensing, and laboratory data from the Provincial Health Data Centre, which includes data from all facilities in the Western Cape province, South Africa (52 hospitals and 354 primary care clinics). We included adults (age 15–85 years) who sought care in Khayelitsha or Gugulethu (which are low-resource, high HIV burden settings), and initiated ART under universal test-and-treat guidelines, implemented from Sept 1, 2016, but before Sept 30, 2021 (to allow >1 year of potential follow-up to data cutoff on Sept 30, 2022). The main outcome was time to first treatment interruption (>90 days late for an expected visit, based on treatment refill data) or censoring (death or data cutoff) after ART initiation and survival analysis explored patterns of recurrent interruption and re-engagement. Cox proportional hazards regression evaluated associations of treatment interruption with demographics (age, sex, area of residence, and access to care), clinical characteristics (comorbidities, pregnancy, hospitalisation, tuberculosis, or COVID-19), and ART history (year of initiation, CD4 cell count, and regimen at initiation).

Findings 68 888 people were included in the cohort, of whom 47 631 (69·1%) were female, 21 189 (30·8%) were male, and 17 078 (24·8%) were aged 25 years or younger, with a median follow-up of 4 years (IQR 3–5). By 6 years of follow-up after ART initiation, cumulative incidence of treatment interruption was 71·8% (95% CI 71·4–72·3), with a median time to interruption of 4·9 months (IQR 0·9–13·0) from initiation. The cumulative incidence of re-engagement after ART interruption at 6 years was 75·6% (95% CI 74·8–76·4), after a median of 7·9 months (IQR 4·4–17·1) after first treatment interruption. Of the 40 384 interruption events with a subsequent return to care, 29 967 (74·2%) events had sufficient follow-up to evaluate outcomes a year after restart, among which 15 062 (50·3%) were virally suppressed (≤1000 copies per mL). The magnitude of association between treatment interruptions and demographic or clinical characteristics was small.

Interpretation Treatment interruptions were common and an expected part of long-term ART use, particularly soon after ART initiation and restart. However, most out of care were individuals at low risk, suggesting the need for more nuanced ways to identify who requires support. Most people returned to care, but ultimately had poor viral suppression, revealing gaps in the way primary care is organised to support sustained viral suppression despite frequent cycling in and out of care.

Funding Clarendon Fund and St John's College.

Lancet Prim Care 2025; 1: 100077

Published Online December 16, 2025
https://doi.org/10.1016/j.lanprc.2025.100077

For the isiXhosa translation of the abstract see Online for appendix 1

Centre for Global Health Research, Nuffield Department of Medicine, University of Oxford, Oxford, UK (C M Keene PhD, Prof M English FMedSci, J McKnight PhD)**; Centre for Infectious Disease Epidemiology and Research, School of Public Health and Family Medicine, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa** (C M Keene, J Euvrard MPH)**; Department of Public Health Medicine, School of Public Health and Family Medicine, University of Cape Town, Cape Town, South Africa** (T Cassidy PhD)**; Department of Medicine, Mass General Brigham Hospital, Boston, MA, USA** (I T Katz MD)**; Desmond Tutu HIV Centre, Institute of Infectious Disease and Molecular Medicine & Department of Medicine, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa** (Prof C Orrell PhD)**; HIV and Other Infectious Diseases Unit, South African Medical Research Council, Durban, South Africa** (Prof C Orrell PhD)

Correspondence to: Dr Claire M Keene, Centre for Global Health Research, Nuffield Department of Medicine, University of Oxford, Oxford, UK **claire.keene@ndm.ox.ac.uk**

Introduction

In South Africa, HIV care is delivered almost entirely through nurse-led primary care clinics and differentiated service delivery models, which serve as the backbone of the public health system and manage the world's largest antiretroviral therapy (ART) programme.[1,2] These clinics operate within a tiered health-care system in which primary care facilities provide first-line chronic disease management, with referral to district and tertiary hospitals for complications or specialist needs. South Africa is making steady progress towards the UNAIDS 95–95–95 targets, but only 75% of the 7·8 million people living with HIV were on antiretroviral therapy (ART) and 69% had a suppressed viral load in 2023.[1] Most (92%) retained on ART were virally suppressed,[1] making retention (ie, engagement with services)[3] the bottleneck to achieving the levels of population-level treatment success required to control the HIV epidemic in South Africa. Supporting long-term retention in HIV care is becoming a priority challenge for ART programmes globally, as disengagement drives the persistence of advanced HIV as well as ongoing transmission.[4] Therefore, understanding patterns of

Research in context

Evidence before this study

Decades into the global HIV response, disengagement from HIV care has become an important challenge as treatment interruptions increase the cost of care and drive transmission of the epidemic. This issue is particularly important in Africa, which is home to two-thirds of people with HIV globally. Cross-sectional methods are widely used to describe engagement with antiretroviral services, but individual-level patterns of care engagement over time, which could provide a more nuanced understanding of this challenge, are rarely reported. We searched PubMed with the search terms (antiretroviral*) AND (retention OR engagement) AND (longitudinal OR "over time"), with no language limits, from Jan 1, 2016, to April 30, 2025. Little evidence is available describing long-term patterns of engagement with care, particularly in services catering to the general population of adults with HIV in low-resource settings, who primarily access HIV services in primary care. Where longitudinal data are available, follow-up tends to be limited to 1–3 years, whereas cohorts with longer follow-up tend to be smaller (<3000 people). There is variability in the incidence of disengagement between settings, with rates as low as 15% in Uganda to 53% in Nigeria over 4 years. Beyond population-level variability, there is little evidence on the timing of risk of disengagement after re-engagement, or long-term retention and treatment outcomes in those who restart after an interruption. What has been found in the few analyses of time to disengagement after restart suggests that the hazard of disengagement is highest soon after re-engagement (within 6 months) and that subsequent treatment outcomes are poor, with low rates of persistent retention and viral suppression. Furthermore, little has been described on how pivotal health events, such as tuberculosis, pregnancy, or hospitalisation, affect engagement with HIV services over time.

Added value of this study

We used a large dataset of routine health-care data to inform a growing body of evidence on longitudinal patterns of engagement in a low-resource, high HIV burden setting. South Africa remains the country with the largest number of people with HIV and the largest antiretroviral programme in the world, provided through primary care clinics and differentiated service delivery models at facility and community sites. This study is particularly valuable as the routine data allowed patterns over 5 years to be evaluated for a large cohort (N=68 888) of people on antiretroviral treatment and provided sufficient observation time to explore long-term patterns of treatment interruptions (including repeat interruptions) and outcomes after re-engagement in this setting. We found that nearly three-quarters of the cohort disengaged at least once within 6 years of initiating antiretroviral treatment, with the greatest hazard of disengagement in the first 6 months. However, longer follow-up revealed greater cumulative attrition than studies with short observation periods. Time to interruption was similar after initiation and restart, regardless of the treatment interruption number. Most people returned to HIV care within 2 years, but subsequent treatment outcomes (ie, viral suppression) were worse in this population than for those who were continuously in care. This study specifically explored the effect of comorbidities and pivotal health events that could affect engagement with HIV services (ie, tuberculosis, pregnancy, hospitalisation, or a COVID-19 diagnosis). We found that comorbidities reduced disengagement, whether present at antiretroviral therapy initiation or developed while on antiretroviral treatment. Having a pivotal health event while on HIV treatment increased the risk of an interruption to care, but starting antiretrovirals during a pivotal health event episode (ie, while under antenatal or tuberculosis care or while hospitalised) had a protective effect that waned after 6 months. We also found that although some demographic and clinical characteristics increased the hazard of a treatment interruption, the magnitude of effect was small and the profile of individuals who had interrupted treatment was similar to those who were continuously in care.

Implications of all the available evidence

This study adds to the evidence that treatment interruptions are very common in this setting: given sufficient follow-up time, most people on antiretroviral therapy will disengage at least once. This finding helps to explain the low cross-sectional estimates of retention in care that, in turn, reduce population-level viral suppression. Considering the detrimental implications of treatment interruptions on individual health and ongoing HIV transmission, support for engagement should remain a priority, even in the context of restricted global funding. Although we found that most people re-engaged with antiretroviral therapy after an interruption, their outcomes were worse than among those who were continuously in care; they would require additional support on return. Health systems need to consider the dynamics of long-term engagement in designing services that acknowledge people dropping out of care as normal, accommodating rather than penalising disengagement, and making it easy for people to return to care as soon as they are able. The protective effect of pivotal health events present at antiretroviral initiation has implications for the way primary care services should be organised and integrated, and provides an opportunity to draw on other services to improve care engagement across the health system. Accurately predicting who will interrupt treatment is not possible using routinely available demographic and clinical characteristics, hence the exploration of more nuanced risk profiles should be a research priority.

disengagement from ART within primary care services is central to improving treatment outcomes and reducing operational burdens for an already stretched health system.

Traditional cross-sectional methods, commonly used to assess HIV cascades (for instance, in a 2023 scoping review,[5] 93% of all observational studies included were cross-sectional studies) have driven global ART implementation and encouraged target-setting to improve population-wide viral suppression,[6] but obscure the population's dynamic interactions with services and treatment, and cannot identify specific timepoints at which people are at risk of disengaging from care.[6] Longitudinal analysis addresses these gaps and is more robust to erroneous inferences than cross-sectional methods.[7] Examining longitudinal population-level engagement with ART therefore complements cross-sectional analyses by providing a nuanced understanding of HIV programme performance and identifying actionable timepoints for interventions to target support.[7]

Few longitudinal studies have examined long-term retention patterns and viral outcomes in primary care services or the effect of repeated treatment interruptions and events (eg, tuberculosis or pregnancy) on engagement, leaving important gaps in understanding the dynamics of long-term retention in HIV care. We aimed to explore longitudinal patterns of disengagement and re-engagement through survival analysis of routine health data from a cohort of people with HIV on ART in the era of universal test-and-treat in the Western Cape, South Africa. With more people now restarting than initiating ART,[8] understanding patterns of retention over time after both initiation and re-engagement is increasingly important for service delivery.

Methods

Study design and data sources

We conducted a retrospective, longitudinal, observational cohort analysis to explore time to treatment interruptions in ART care through survival analysis, following STROBE reporting guidelines.[9] The protocol is available on the Oxford University Research Archive, and ethical approval for this study was obtained from the University of Cape Town's Human Research Ethics Committee (066/2022).[9] Individual patient consent was not obtained, as data was anonymised on export from the routine data centre.

Khayelitsha and Gugulethu are two periurban settlements outside Cape Town in the Western Cape province of South Africa, with a total estimated population of 1 million people. Both are characterised by high HIV prevalence and large, long-established cohorts on ART, against a backdrop of high rates of non-communicable disease, poverty, unemployment, violence, and high mobility for work, making engagement with HIV care particularly challenging.[10] Routine ART dispensing is generally every 1–2 months, through primary care clinics or differentiated models, such as facility-based or community-based adherence clubs.

Demographic (including sex, based on patient records), mortality, dispensing, and laboratory data from across all facilities in the Western Cape (52 hospitals and 354 primary care clinics) were obtained from the Provincial Health Data Centre (PHDC; appendix 2 p 2). The PHDC is a health-data repository of linked, individual-level information for an estimated 8 million patients (>430 000 of whom have HIV) from existing governmental routine health data systems within the province, consolidated and curated by the Western Cape Department of Health and Wellness.[11]

Data were provided starting from Jan 1, 2013, and data across all facilities in the province were included for adults (age 15–85 years) starting ART between Sept 1, 2016, (ie, the date of implementation of universal test-and-treat guidance in the Western Cape to create a more homogenous eligibility for ART) and Sept 30, 2021, with data available (including pharmacy refill data) for at least 1 year of follow up until data cutoff on Sept 30, 2022, and seeking care at some point in Khayelitsha or Gugulethu (appendix 2 pp 3–4).

Procedures

Variables with potential to influence engagement were demographics (age, sex, area of residence, and access to care) and ART history (year of initiation, CD4 cell count, and regimen at initiation, to account for the introduction of dolutegravir as first-line ART in 2020). Clinical characteristics comprised comorbidities (diabetes, hypertension, or a mental health diagnosis) and pivotal health events that could affect engagement (ie, pregnancy, hospitalisation, and a tuberculosis or COVID-19 diagnosis). Variables were selected from those available in the routine data, recorded in the PHDC from clinical and administrative records (appendix 2 pp 2, 6). Data on race and ethnicity were not available from the PHDC.

Time to a substantial treatment interruption was defined as being more than 90 days late for an expected visit based on pharmacy refill data or being lost to follow-up (no visit within 180 days of data cutoff), as this provides a granular measure of people outside of care, strongly associated with viral outcomes,[12] and is the threshold used in the 2023 South African National Department of Health Antiretroviral Therapy Guidelines to direct the management of re-engaging people with HIV with health care.[2] Treatment interruptions were counted as starting on the expected visit date. The frequency of treatment interruptions that were longer than 7 days, longer than 30 days, and longer than 42 days was also explored descriptively.

For the **protocol** see https://ora.ox.ac.uk/objects/uuid:b8d73cf3-a6e0-4ae0-ac54-f36016397366

1-year retention was defined as an ART refill visit 9–15 months after ART initiation or restart. Viral suppression was defined as a viral load of 1000 copies per mL or less and undetectable viral load was defined as 50 copies per mL or less, as per WHO guidance,[13] but included the threshold value because viral loads were rounded on data export to preserve confidentiality (appendix 2 p 5). A 12-month window (6 months before and after the expected viral load due date) was used to account for flexibility in when guideline-stipulated yearly viral loads were actually measured.[2]

Statistical analysis

See Online for appendix 2

We described characteristics with counts and proportions, and continuous variables with medians and IQRs (non-normally distributed data) and present these for the whole cohort and stratified by those continuously in care (no treatment interruption) and not continuously in care (one or more treatment interruptions during follow-up). Treatment outcomes were described 1 year after starting ART (initiation or restart) in those with sufficient follow-up.

We did a Kaplan–Meier survival analysis for the description of time to treatment interruption or re-engagement, presented as cumulative incidence. The primary analysis was time to first treatment interruption, measured from ART initiation until the first interruption or censoring. We did post-hoc exploratory analyses to investigate time to recurrent interruption events (re-zeroing the start time on ART restart and evaluating time until the next treatment interruption or censoring) and time to re-engagement (measuring time from the expected visit date to ART restart) to describe subsequent interruptions and re-engagement at the episode level. All analyses right censored individuals at death or data cutoff (Sept 30, 2022), whichever came first.

We used an extended Cox proportional hazards model to assess the association of variables with time to the first treatment interruption.[14] Proportional hazards assumptions were tested graphically with Schoenfeld residuals for time-independent variables and log cumulative hazard plots for variables appearing non-proportional.[14] For pivotal health events at ART initiation, we used log cumulative hazard plots to inform the inclusion of interaction terms with follow-up time in the extended Cox model. The variance inflation factor assessed multicollinearity of candidate variables, accommodating categorical variables through dummy expansion, and with a variance inflation factor of 3 or more interpreted as a conservative screening indicator (due to the large sample) of potential collinearity warranting inspection (rather than exclusion).[15]

Demographics, ART history, and the presence of comorbidities at ART initiation were considered to be baseline, time-independent variables. New comorbidities and pivotal health events that occurred after ART initiation were considered inherently time dependent. The effect of pivotal events present at initiation did not meet the proportional hazards assumption, so we included a time-by-covariate interaction term (at 6 months). A post-hoc exploratory sensitivity analysis examined no interaction term, an interaction term at 165 days after initiation (crossover of log cumulative hazards), and stratifying by pivotal health event at initiation. The COVID-19 pandemic was not experienced equitably and occurred at different points in each individual's follow-up after initiation, so was included in the multivariable model to adjust for this effect. All described variables were evaluated with univariable analysis. Multivariable analysis identified the final model through a backwards selection strategy: variables with the highest p value were removed one by one until all remaining variables had a statistically significant contribution to the outcome ($p<0{\cdot}05$, to account for large sample size).[16] Model fit was reassessed after each step with likelihood-ratio tests and the Akaike Information Criterion, ensuring that model simplification did not worsen fit.

The primary analysis of viral load outcomes considered a missing viral load as not virally suppressed. A prespecified sensitivity analysis included only those with a completed viral load and those retained in care at data cutoff with a completed viral load, to account for individuals who might not have a viral load measured or documented, particularly during the COVID-19 pandemic.

The analysis was done with R, version 4.3.2, tidyverse, survival, condsurv, and bshazard packages, through the RStudio interface.

Role of the funding source

The funder had no role in the study design, data collection, data analysis, data interpretation, or writing of the report.

Results

A cohort of 68 888 adults started ART during the universal test-and-treat era in Khayelitsha and Gugulethu and were eligible for inclusion in our analysis (figure 1). Missing data are reported in appendix 2 (pp 7–8). Median follow-up was 4 years (IQR 3–5 years).

In the whole cohort, 47 631 (69·1%) of 68 888 people were female, 21 189 (30·8%) were male, the median age was 31 years (IQR 26–38), and 51 810 (75·2%) were older than 25 years. Regardless of continuity of care, age distributions and female to male proportions were similar. Most people accessed care in one region, did not have a comorbid diagnosis, were on non-nucleoside reverse transcriptase inhibitor -based regimens at initiation, and had an initial CD4 count with a median of 300 cells per µL (IQR 200–500; table 1). Those not continuously in care were slightly younger, had fewer chronic disease diagnoses, and slightly longer follow-up observation time, and a smaller proportion were on integrase strand transfer inhibitor (INSTI)-based regimens, both at initiation and at censoring (dolutegravir was scaled-up in 2020, so differences at initiation are likely confounded by earlier ART start, and at data cutoff because people were out of care and not available to be switched) than those in continuous care. The proportion of individuals who died was higher among those continuously retained in care than in those with interruptions (table 1), potentially reflecting a data bias in which those who died while out of care were not captured. Appendix 2 (pp 9–10) provides further information on specific comorbidities and pivotal health events, stratified by timing relative to ART initiation.

In the last year of follow-up, 33 377 (48·5%) of 68 888 participants were virally suppressed, with 20 091 (85·2%) of 23 578 participants who were continuously in care suppressed compared with 13 286 (29·3%) of 45 310 who ever had interrupted treatment during follow-up. Differences remained when examining only those retained in care with a viral load result (appendix 2 p 11). Only 4775 (15·0%) of 31 883 people missing a viral load in their last year of follow-up were retained in care at censoring, compared with 1438 (60·9%) of 2360 people who had

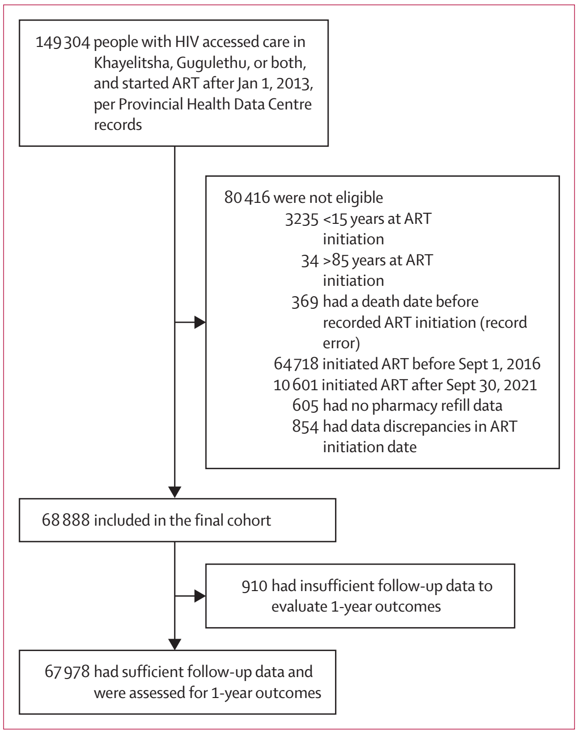


Figure 1: **Flow diagram of cohort eligibility**
ART=antiretroviral therapy.

an unsuppressed viral load who were retained in care (appendix 2 p 11).

Disengagement with care was common across the cohort. 45 310 (65·8%) of 68 888 people had at least one substantial treatment interruption (>90 days late) at some point in their follow-up. The median number of treatment interruptions was 1 (IQR 1–2; range 1–10). Additionally, 65 610 (95·2%) of 68 888 people had treatment interruptions of more than 7 days, 59 401 (86·2%) of 68 888 by more than 30 days, and 56 953 (82·7%) of 68 888 by more than 42 days. There was a median of 1 shorter interruption (IQR 0–2) of 7–90 days before each substantial treatment interruption of more than 90 days.

The cumulative incidence of treatment interruption by 6 years after ART initiation was 71·8% (95% CI 71·4–72·3; figure 2A), with 12 092 (18%) of 68 888 interrupting after the first month and 24 960 (36%) in the first 6 months of treatment. Similar patterns were found for time to interruption from ART initiation and restart for the first five interruptions (figure 2B).

In those with treatment interruptions, the median time to interruption was 4·9 months (IQR 0·9–13·0) from initiation and 3·7 months (IQR 0·9–8·6) for recurrent treatment interruptions after restart. The hazard of an interruption was highest soon after ART initiation (ie, within the first 6 months) and dropped precipitously over the first 6–12 months, with a similar pattern but greater magnitude for time to recurrent interruptions (appendix 2 p 12).

Univariable and multivariable analyses of associations between demographic and clinical characteristics and ART

	Whole cohort (N=68 888)	Continuously in care* (n=23 578)	Not continuously in care† (n=45 310)
Demographics			
Age at initiation, years	31 (26–38)	33 (27–41)	30 (25–37)
Age group at initiation, years			
15–25	17 078 (24·8%)	4197 (17·8%)	12 881 (28·4%)
26–85	51 810 (75·2%)	19 381 (82·2%)	32 429 (71·6%)
Sex			
Female	47 631 (69·1%)	16 486 (69·9%)	31 145 (68·7%)
Male	21 189 (30·8%)	7067 (30·0%)	14 122 (31·2%)
Unknown	68 (0·1%)	25 (0·1%)	43 (0·1%)
Location of care accessed			
Khayelitsha only	36 823 (53·5%)	12 736 (54·0%)	24 087 (53·2%)
Gugulethu only	28 660 (41·6%)	9952 (42·2%)	18 708 (41·3%)
Both	3405 (4·9%)	890 (3·8%)	2515 (5·6%)
Residence			
Khayelitsha only	28 978 (42·1%)	10 124 (42·9%)	18 854 (41·6%)
Gugulethu only	15 509 (22·5%)	5482 (23·3%)	10 027 (22·1%)
Both‡	607 (0·9%)	194 (0·8%)	413 (0·9%)
Neither	23 794 (34·5%)	7778 (33·0%)	16 016 (35·3%)
Clinical history			
Chronic disease diagnoses at initiation	7062 (10·3%)	3343 (14·2%)	3719 (8·2%)
Hypertension	5177 (7·5%)	2617 (11·1%)	2560 (5·6%)
Diabetes	1556 (2·3%)	787 (3·3%)	769 (1·7%)
Mental health diagnosis	1537 (2·2%)	567 (2·4%)	970 (2·1%)
Pivotal health events at initiation	17 396 (25·3%)	5294 (22·5%)	12 102 (26·7%)
Hospital admission	1313 (1·9%)	344 (1·5%)	969 (2·1%)
COVID-19 diagnosis	83 (0·1%)	41 (0·2%)	42 (0·1%)
Tuberculosis	7784 (11·3%)	2638 (11·2%)	5146 (11·4%)
Pregnancy as a proportion of females	8991 (18·9%)	2488 (15·1%)	6503 (20·9%)
HIV history			
Observation time since ART initiation, years	3·99 (2·75–5·08)	3·58 (2·17–4·84)	4·18 (3·01–5·18)
ART initiation, year	2018 (2017–19)	2019 (2017–20)	2018 (2017–19)
Regimen at ART initiation			
NNRTI based	53 914 (78·3%)	16 959 (71·9%)	36 955 (81·6%)
PI based	790 (1·1%)	134 (0·6%)	656 (1·4%)
INSTI based	12 515 (18·2%)	6189 (26·2%)	6326 (14·0%)
Unknown	1669 (2·4%)	296 (1·3%)	1373 (3·0%)
CD4 cell count at initiation	45 698 (66·3%)	16 987 (72·0%)	28 711 (63·4%)
CD4 cell count, cells per µL	300 (200–500)	300 (200–500)	300 (200–500)
Clinical events after ART initiation			
Death	1957 (2·8%)	1132 (4·8%)	825 (1·8%)
Chronic disease diagnoses	6278 (9·1%)	2614 (11·1%)	3664 (8·1%)
Hypertension	3856 (5·6%)	1759 (7·5%)	2097 (4·6%)
Diabetes	974 (1·4%)	435 (1·8%)	539 (1·2%)
Mental health diagnosis	1876 (2·7%)	580 (2·5%)	1296 (2·9%)
Pivotal health events	27 824 (40·4%)	8958 (38·0%)	18 866 (41·6%)
Hospital admission	16 260 (23·6%)	5230 (22·2%)	11 030 (24·3%)
COVID-19 diagnosis	2704 (3·9%)	1054 (4·5%)	1650 (3·6%)
Tuberculosis	5820 (8·5%)	1404 (6·0%)	4416 (9·7%)
Pregnancy as a proportion of females	13 482 (28·3%)	4274 (25·9%)	9208 (29·6%)

(Table 1 continues on next page)

	Whole cohort (N=68 888)	Continuously in care* (n=23 578)	Not continuously in care† (n=45 310)
(Continued from previous page)			
ART regimen at censoring			
NNRTI-based	23 108 (33·5%)	2140 (9·1%)	20 968 (46·3%)
PI-based	1382 (2·0%)	324 (1·4%)	1058 (2·3%)
INSTI-based	42 608 (61·9%)	20 828 (88·3%)	21 780 (48·1%)
Unknown	1790 (2·6%)	286 (1·2%)	1504 (3·3%)

Data are median (IQR) or n (%). ART=antiretroviral therapy. INSTI=integrase strand transfer inhibitor. NNRTI=non-nucleoside reverse transcriptase inhibitor. PI=protease inhibitor. *Did not have a treatment interruption during follow-up. †Treatment interruption at some point during follow-up. ‡During follow-up, people could have registered addresses in both settings.

Table 1: **Cohort characteristics**

history with time to treatment interruption after ART initiation are shown in table 2. A post-hoc sensitivity analysis found similar hazard ratios using no interaction term, an interaction term at 165 days after initiation (crossover of log cumulative hazards), and stratifying by pivotal health event at initiation (appendix 2 pp 17–18).

Among those who had interrupted care, the majority eventually returned to care within 2 years (appendix 2 p 19). The cumulative incidence of re-engagement after ART interruption was 75·6% (95% CI 74·8–76·4) by 6 years after an interruption (appendix 2 p 19). Median time to re-engagement was 7·9 months (IQR 4·4–17·1) for those who returned from their first treatment interruption and 6·3 months (4·1–11·8) for those returning from subsequent treatment interruptions.

27 734 individuals had a substantial treatment interruption and returned to care before the end of the study, among whom there were a total of 40 384 treatment interruption events. Considering each treatment interruption independently, 29 967 (74·2%) interruption events had sufficient follow-up to evaluate outcomes 1 year after re-engagement. In 20 030 (66·8%) interruptions, the individual was retained a year after re-engagement and in 15 062 (50·3%) they had a suppressed viral load (table 3). In comparison, in those who were continuously in care, 18 569 (81·6%) of 22 768 were virally suppressed 1 year after initiation (table 3).

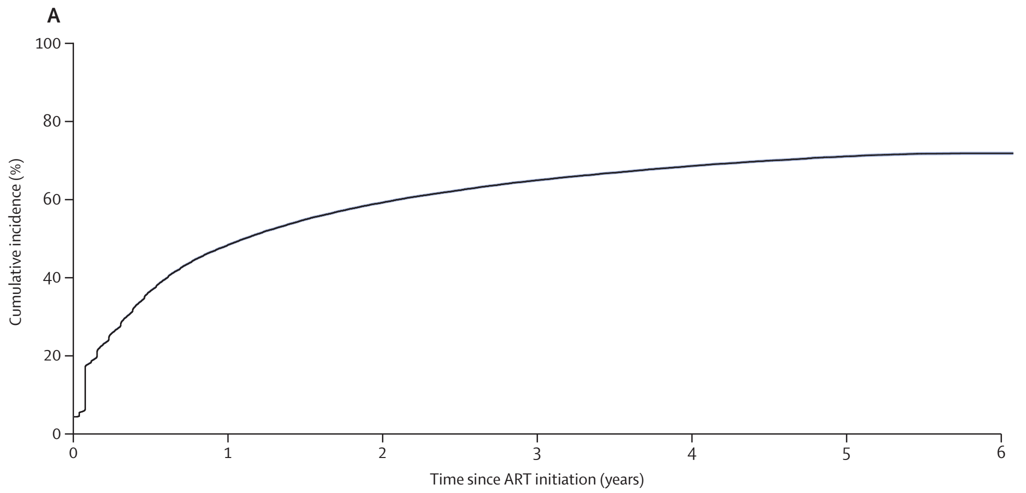
Sensitivity analyses of retention and suppression by sequential interruption number and for different viral load thresholds are presented for each definition of suppression in the appendix 2 (pp 20–22). Those continuously in care had greater viral suppression 1 year after initiation than those with an interruption 1 year after ART restart, regardless of interruption number, definition, or threshold for suppression.

Discussion

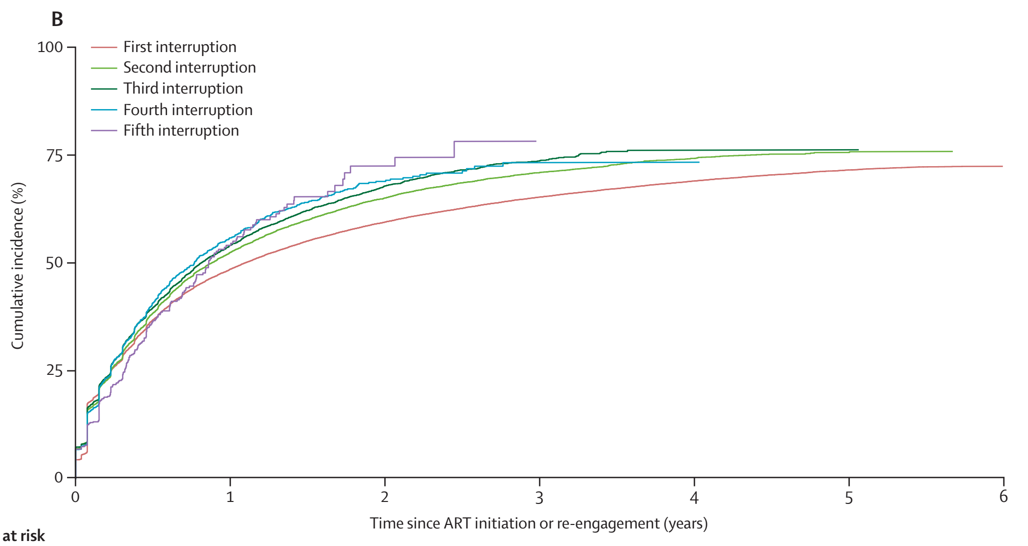
Longitudinal patterns of care interruption and re-engagement complement cross-sectional snapshots, revealing cycling of people with HIV coming in and out of care. Interruptions were common, particularly in the first 6–12 months of ART, but the majority re-engaged within 2 years. Retention patterns after restart resembled those after initiation, albeit with a higher hazard of interruption, and substantially worse treatment outcomes after re-engagement. Demographic and clinical profiles of people who had interrupted ART were similar to those continuously in care, with small associations between hazard of interruptions and some demographic, clinical, and ART history characteristics.

Given sufficient follow-up time, substantial (>90 days) interruptions appear inevitable: nearly three-quarters were expected to interrupt treatment by 6 years on ART. Shorter interruptions were very common, with 95% of the cohort recorded as late for an expected appointment by more than 7 days at least once during follow-up. Cumulative attrition was worse than other cohorts with shorter observation periods, which ranges from 15% in Uganda[17] to 53% in Nigeria[18] over 4 years. Although most people had only one interruption, approximately a third of those with a gap in care had multiple interruptions. The pattern was similar across sequential treatment interruptions; the hazard of disengagement was highest in the first 6 months after both initiating and restarting ART, echoing widely published data emphasising this high-risk period after initiation across different contexts.[19] Our study suggests that prioritisation of the first 6 months should be extended to people returning to care, a group likely to form an increasing proportion of those starting treatment in the future.[8]

Although the pattern of interruptions was similar after restart and initiation, outcomes were not. Retention and viral suppression were lower after re-engagement, reinforcing evidence that suppression is associated with retention continuity.[20] Many of the factors contributing to previous disengagement could still be present on return, meaning that these people might restart ART at an already greater risk of interruption.[20] These findings demonstrate the “revolving door of HIV care” described by Ehrenkranz and colleagues:[21] most patients disengage at least once (often early) in their treatment journey, but most re-engage within 2 years. Even on return, lower rates of viral suppression below the UNAIDS target of 95% suggest people tend to struggle with adherence.[3] Most people with HIV will probably need engagement support at some point, if not multiple times,[20] suggesting that it is normal to struggle with engagement.[22] This pattern does not imply that interruptions should be ignored: substantial interruptions increase risk of viraemia, viral failure, drug resistance, and death, threatening the control of the HIV epidemic.[23,24] But this pattern does require a paradigm shift. Services as they are currently structured do not consistently work for most people throughout their life (as evidenced by frequent disengagement) and are not flexible enough to accommodate people with HIV moving in and out of care. For example, bureaucratic requirements for transfer letters when changing facilities impede people’s ability to self-manage and remain engaged despite challenges in their complex lives; even if they want to re-engage elsewhere, they are often turned away.[25] Primary care service design and intervention implementation need to reflect this lived reality.



	0	1	2	3	4	5	6
Number at risk (number censored)	68 888 (0)	34 938 (845)	23 629 (5284)	16 659 (9178)	10 568 (13 770)	5235 (18 447)	370 (23 208)
Cumulative incidence, % (95% CI)	0%	48·4% (48·0–48·8)	59·2% (58·9–59·6)	65·0% (64·6–65·3)	68·6% (68·2–69·0)	71·1% (70·7–71·5)	71·8% (71·4–72·3)



Number at risk (number censored)	0	1	2	3	4	5	6
First interruption	68 888 (0)	34 938 (845)	23 629 (5284)	16 659 (9178)	10 568 (13 770)	5235 (18 447)	370 (23 208)
Second interruption	27 734 (0)	9539 (4991)	4797 (7580)	2437 (9286)	805 (10 733)	147 (11 365)	0 (11 511)
Third interruption	9052 (0)	2351 (2533)	837 (3516)	299 (3941)	44 (4177)	1 (4220)	0 (4221)
Fourth interruption	2696 (0)	498 (1028)	120 (1304)	23 (1391)	1 (1413)	0 (1414)	0 (1414)
Fifth interruption	702 (0)	98 (349)	15 (409)	0 (422)	0 (422)	0 (422)	0 (422)

Figure 2: **Cumulative incidence of treatment interruption**
(A) Cumulative incidence of time to the first treatment interruption of more than 90 days after an expected visit, from ART initiation (95% CI in blue). (B) Cumulative incidence of time to treatment interruption from ART initiation or restart, stratified by the sequential treatment interruption number for the first five interruptions. 95% CI omitted for part B to aid visualisation. ART=antiretroviral therapy.

	Univariable analysis		Multivariable analysis	
	Hazard ratio (95% CI)	p value	Adjusted hazard ratio (95% CI)	p value
Demographics				
Age at ART initiation				
>25 years	1 (ref)	··	1 (ref)	··
<25 years	1·43 (1·40–1·46)	<0·0001	1·25 (1·22–1·28)	<0·0001
Sex				
Female	1 (ref)	··	1 (ref)	··
Male	1·05 (1·02–1·07)	<0·0001	1·20 (1·18–1·23)	<0·0001
Care location				
Both	1 (ref)	··	1 (ref)	··
Khayelitsha	0·85 (0·81–0·88)	<0·0001	0·95 (0·92–0·99)	0·023
Gugulethu	0·84 (0·81–0·88)	<0·0001	0·92 (0·88–0·96)	<0·0001
Residence location				
Both Khayelitsha and Gugulethu	1 (ref)	··	NA*	··
Khayelitsha only	0·92 (0·83–1·01)	0·091	NA*	··
Gugulethu only	0·91 (0·82–1·00)	0·054	NA*	··
Neither Khayelitsha nor Gugulethu	0·96 (0·87–1·06)	0·40	NA*	··
Clinical history				
Comorbidity at ART initiation				
No comorbidity	1 (ref)	··	1 (ref)	··
Has comorbidity	0·66 (0·64–0·68)	<0·0001	0·79 (0·76–0·81)	<0·0001
Pivotal health event at ART initiation				
No pivotal event	1 (ref)	··	1 (ref)	··
Pivotal event in the first 180 days	0·72 (0·67–0·77)	<0·0001	0·76 (0·70–0·82)	<0·0001
Pivotal event after 180 days	1·35 (1·31–1·42)	<0·0001	1·41 (1·37–1·46)	<0·0001
HIV history				
ART initiation year	0·99 (0·99–1·00)	0·073	0·86 (0·85–0·87)	<0·0001
CD4 at ART initiation				
CD4 >200	1 (ref)	··	1 (ref)	··
Advanced HIV disease (CD4 ≤200)	0·94 (0·92–0·96)	<0·0001	0·93 (0·91–0·96)	<0·0001
No CD4 at baseline	1·28 (1·26–1·31)	<0·0001	1·27 (1·25–1·30)	<0·0001
Regimen at ART initiation				
NNRTI based	1 (ref)	··	1 (ref)	··
PI based	1·98 (1·84–2·14)	<0·0001	1·61 (1·48–1·75)	<0·0001
INSTI based	0·79 (0·77–0·81)	<0·0001	0·91 (0·88–0·94)	<0·0001
Unknown	1·70 (1·61–1·80)	<0·0001	1·44 (1·36–1·53)	<0·0001
Clinical events after ART initiation				
Comorbidity diagnosed after initiation				
No comorbidity diagnosed	1 (ref)	··	1 (ref)	··
Comorbidity	1·01 (0·96–1·06)	0·73	0·66 (0·64–0·69)	<0·0001
Pivotal event after initiation				
No pivotal event	1 (ref)	··	1 (ref)	··
Has pivotal event	1·32 (1·29–1·35)	<0·0001	1·22 (1·19–1·25)	<0·0001
COVID-19 pandemic				
Time in the COVID-19 pandemic				
Time pre or post pandemic	1 (ref)	··	1 (ref)	··
Time in the pandemic pre-outcome	1·13 (1·11–1·16)	<0·0001	1·76 (1·71–1·80)	<0·0001

Data are from Cox proportional hazards regression models. Assessment for Schoenfeld residuals and log cumulative hazard plots are in appendix 2. All variables were evaluated in the unadjusted model. Variables retained in the adjusted multivariable model were those that met statistical criteria for inclusion (p<0·05) during backwards selection. All retained variables were adjusted for each other. *Removed in backwards selection. Variance Inflation Factor values of all variables were close to 1, indicating minimal multicollinearity (appendix 2). ART=antiretroviral therapy. INSTI=integrase strand transfer inhibitor. NNRTI=non-nucleoside reverse transcriptase inhibitor. PI=protease inhibitor.

Table 2: **Association of demographic, clinical, and engagement variables with engagement, measured as the time to the first treatment interruption after ART initiation**

	Outcome 1 year after initiation		Outcome 1 year after restart for all treatment interruption events (n=40 384)*
	Whole cohort (N=68 888)	Continuously in care (n=23 578)	
Sufficient follow-up to assess 1-year outcomes	67 978 (98·7%)	22 768 (96·6%)	29 967 (74·2%)
Virally suppressed	34 545/67 978 (50·8%)	18 569/22 768 (81·6%)	15 062/29 967 (50·3%)
Viral load done (as a proportion of those with sufficient follow-up)	37 564/67 978 (55·3%)	19 205/22 768 (84·4%)	17 244/29 967 (57·5%)
Virally suppressed	34 545/37 564 (92·0%)	18 569/19 205 (96·7%)	15 062/17 244 (87·3%)
Retained in care† (as a proportion of those with sufficient follow-up)	45 179/67 978 (66·5%)	22 613/22 768 (99·3%)‡	20 030/29 967 (66·8%)
Viral load done	35 872/45 179 (79·4%)	19 205/22 613 (84·9%)	15 568/20 030 (77·7%)
Virally suppressed	33 377/35 872 (93·0%)	18 569/19 205 (96·7%)	13 676/15 568 (87·8%)

Data are n (%) or n/N (%). Viral load results are within a 12-month window (6 months either side of the fixed point of 1 year after initiation or restart) and viral suppression is defined as a viral load of maximum 1000 copies per mL. *Data for this column are for the number of events rather than number of people. †Retention in care calculated with fixed point retention at each timepoint, with a window of 180 days (90 days either side of the fixed point). ‡This is not 100% due to different definitions of treatment interruptions and being retained in care: 1-year retention was defined as a refill visit 9–15 months after initiation or restart, whereas treatment interruptions were defined as being more than 90 days late for an expected visit based on refill duration. Depending on the duration of the antiretroviral refill, individuals might not have had a visit around the 1-year timepoint while not having interrupted treatment for more than 90 days. Therefore, despite no treatment interruption, they would not have a 1-year viral load completed.

Table 3: **Retention and viral load outcomes 1 year after antiretroviral initiation or restart after re-engagement**

Rather than expecting people with HIV to adapt to vertical systems developed to scale access to treatment in a previous era of the epidemic, and penalising them when they do not conform, primary care services need to expedite adoption of person-centred care principles to respond to how people actually behave and provide what they need to remain engaged long term.[26] If interrupting treatment is normal, implementation approaches that accommodate and mitigate the effect of interruptions when they inevitably occur should have a larger role.

People at risk of interruptions probably need different services to support their re-engagement, sustained retention, and long-term viral suppression, compared with those continuously retained and coping with the system as it is currently structured.[20] The factors associated with higher hazards of treatment interruptions were as expected from the wider literature:[27] being male, being younger than 25 years, moving care location, or having a pivotal health event after ART initiation all increased the hazard of interruption. These demographic and clinical characteristics have traditionally been targeted for intervention, such as male-focused or youth-focused clinical services or additional follow-up for pregnant women and people with tuberculosis, resulting in greater improvements in these subpopulations.[27] However, the effect sizes were modest and not strongly predictive; no adjusted HR increased risk by even double the reference. Rather, the population who had interruptions appeared similar in measured variables to those who were continuously retained throughout their follow-up: most were females in their 20s and 30s, with similar rates of chronic disease and pivotal health events as those in continuous care. These data suggest that demographics and clinical histories are not good predictors of engagement behaviour and that other ways to stratify the population should be explored.

The protective effect of comorbidities or pivotal health events at ART initiation suggests that individuals already in care for other reasons are more likely to engage with HIV programmes. This finding might reflect selection for people already engaging with health care, or could point to experience with services, such as outpatient chronic disease clinics for hypertension or diabetes, increasing a person with HIV's capability, opportunity, or motivation to engage. Events with the potential to have a pivotal effect on engagement (ie, pregnancy, tuberculosis, hospitalisation, or COVID-19 diagnosis) increased the risk of disengagement, as would be expected from current literature.[28,29] However, initiating ART during a pivotal event episode initially had a protective effect that waned after 6 months. Pivotal events provide the health system with an opportunity to identify people with HIV and initiate or restart ART. Tuberculosis and antenatal services in particular put great effort into ensuring their patients are on ART due to associated risks of morbidity and vertical transmission, respectively. After a period of intense follow-up and special interest in their engagement under these specialised services, individuals move back to routine ART services, at which point they are at higher hazard of interrupting treatment, according to our results. The impact of each pivotal health event separately and the drivers of this initial protection both warrant further exploration, to harness these mechanisms to improve support and integration throughout primary care services.

Most longitudinal analyses of engagement in low-resource, high HIV burden settings examine interruptions over 1–2 years of follow-up. The much longer observation period in this large cohort allows exploration of long-term patterns of disengagement after ART initiation in a low-resource context and provides enough follow-up to meaningfully explore patterns of interruptions and long-term outcomes after re-engagement. However, more than a quarter of interruptions had insufficient follow-up to evaluate 1-year outcomes, and although data linkage accounts for local so-called silent transfers between facilities in the province, interactions with services outside of the Western Cape could not be assessed. Treatment interruptions adjust for different durations of pharmacy refills over time but assume that individuals take dispensed treatment until it is finished (expected visit date), which might not always reflect the reality of individual behaviour. Although, by definition, treatment interruptions account for changes in refill duration in response to the COVID-19 pandemic, some missing viral load results could be explained by shifting priorities during the pandemic. Viral load results were also rounded off on data export to reduce re-identification risk, which could affect suppression rates.

See Online for appendix 3

These findings are probably generalisable to similar public-sector ART programmes in southern Africa with long-standing treatment cohorts and mobile populations. However, results should be interpreted carefully given the observational design in only two specific geographical areas in one province of South Africa, reliance on imperfect routine data, and the exploratory nature of the time to recurrent events and re-engagement analyses. Additionally, we did not account for competing risks, since death was uncommon and transfer out status was not available. Although differences between censored and uncensored individuals were observed (data not shown), most censoring reflected database closure. Transfers could have also been misclassified as being out of care. The association of demographics and clinical characteristics was not evaluated for time to repeat events, race/ethnicity were not available for analysis, and pivotal events were not evaluated independently; future work could explore this.

Treatment interruptions were common in people with HIV on treatment in this setting, underscoring the need to reframe them as an expected challenge in a lifetime on ART. Although most people re-engaged with care during study follow-up, their worse retention and treatment outcomes than those continuously in care highlight ongoing risk and persistent gaps in current primary care service models. Traditional demographic and clinical characteristics were not strongly predictive of disengagement, and people with HIV with interruptions had similar characteristics to those continuously in care, revealing that routine data is insufficient to identify people at risk, both in a clinical consultation and for analyses to influence policy. There is a need for further research to identify more nuanced markers of interruption risk and develop policy on person-centred approaches that anticipate and support people with HIV when they disengage. Given that HIV care in much of sub-Saharan Africa is delivered through primary care, these services need to evolve to accommodate competing priorities and pivotal health events that disrupt engagement, integrating HIV and non-communicable disease care, supporting transitions from other services (such as tuberculosis or antenatal care), and ensuring that re-engagement is easy, non-punitive, and supportive across the network of primary care clinics to sustain long-term viral suppression in people living with HIV.

Contributors

Conceptualisation: CMK, CO, ITK, JMcK, and ME. Methodology: CMK, JE, TC, and CO. Analysis: CMK and reviewed by JE, TC, and CO. Writing: CMK wrote the original draft, which was reviewed and edited by CO, JE, TC, ITK, JMcK, and ME. CMK and JE both directly accessed and verified the underlying data. All authors had the ability to access to all the data in the study and had final responsibility for the decision to submit for publication.

Equitable partnership declaration

The authors of this paper have submitted an equitable partnership declaration (appendix 3). This statement allows researchers to describe how their work engages with researchers, communities, and environments in the countries of study. This statement is part of *The Lancet Primary Care*'s broader goal to decolonise global health.

Declaration of interests

TC was employed by Gilead Sciences at the time of analysis; Gilead Sciences had no involvement in the design, conduct, analysis, interpretation, or reporting of this study. All other authors declare no competing interests.

Data sharing

Access to data is through application directly to the Western Cape Provincial Health Data Centre. Information on the approval process can be found at https://d7.westerncape.gov.za/general-publication/health-research-approval-process.

Acknowledgments

This research was done as part of a PhD by CMK at the University of Oxford (Oxford, UK). It was supported through a scholarship from the Clarendon Fund and St John's College Kendrew Clarendon Award, in partnership with the Nuffield Department of Clinical Medicine. We would like to thank Cherry Lim, Paul Mwaniki, the Primary Health Care Stats Coven at the University of Oxford, Jason Oke, and Adrian Smith for their input on this analysis as part of their support for CMK's PhD. Thank you also to the Provincial Health Data Centre team, and Bhekumusa Nombela in particular, for their patience and effort to process the data, for their feedback on these analyses, and for all the work they do to support care in the Western Cape. Finally, thank you to Nompumelelo Zokufa and Keitumetse Lebelo for translating the abstract into isiXhosa. During the preparation of this work the authors used ChatGPT (OpenAI, GPT-4, 2023) to assist with refining code during data analysis and ChatGPT (OpenAI, GPT-5, 2025) for language editing, clarifying reviewer responses, and improving phrasing for clarity and concision. After using this tool, the authors reviewed and verified all AI-generated content and take full responsibility for the final manuscript. No AI tools were used in the research design or data interpretation.

References

1 UNAIDS. Country factsheets: South Africa 2024. 2025. https://www.unaids.org/en/regionscountries/countries/southafrica (accessed Aug 5, 2025).
2 South African National Department of Health. 2023 ART clinical guidelines for the management of HIV in adults, pregnancy and breastfeeding, adolescents, children, infants and neonates. 2023. https://knowledgehub.health.gov.za/system/files/elibdownloads/2023-05/National%20ART%20Clinical%20Guideline%202023_04_28%20signed_0.pdf (accessed Dec 9, 2025).
3 Keene CM, Euvrard J, Amico KR, et al, and the InCARE Stakeholder Group. Conceptualising engagement with HIV care for people on treatment: the Indicators of HIV Care and Antiretroviral Engagement (InCARE) Framework. *BMC Health Serv Res* 2023; **23:** 435.
4 Jarvis JN, Ehrenkranz P, Lawrence DS, et al. Reducing HIV incidence and mortality: two sides of the same coin in the approach to ending AIDS. *Lancet HIV* 2025; published online Oct 21. https://doi.org/10.1016/S2352-3018(25)00265-6.
5 Kalinjuma AV, Glass TR, Masanja H, et al. Statistical methods applied for the assessment of the HIV cascade and continuum of care: a systematic scoping review. *BMJ Open* 2023; **13:** e071392.

6 Powers KA, Miller WC. Critical Review: building on the HIV cascade: a complementary "HIV states and transitions" framework for describing HIV diagnosis, care, and treatment at the population level. *J Acquir Immune Defic Syndr* 2015; **69**: 341–47.

7 Haber NA, Lesko CR, Fox MP, et al. Limitations of the UNAIDS 90–90–90 metrics: a simulation-based comparison of cross-sectional and longitudinal metrics for the HIV care continuum. *AIDS* 2020; **34:** 1047–55.

8 Euvrard J, Timmerman V, Keene CM, et al. The cyclical cascade of HIV care: temporal care engagement trends within a population-wide cohort. *PLoS Med* 2024; **21:** e1004407.

9 von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP, and the STROBE Initiative. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol* 2008; **61:** 344–49.

10 South African Strategic Development Information and GIS Department. 2011 census suburb profiles. 2013. https://resource.capetown.gov.za/documentcentre/Documents/Maps%20and%20statistics/2011_Census_CT_Suburb_Observatory_Profile.pdf (accessed Dec 9, 2025).

11 Boulle A, Heekes A, Tiffin N, et al. Data centre profile: the Provincial Health Data Centre of the Western Cape. *Int J Popul Data Sci* 2019; **4:** 1143.

12 Keene CM, Euvrard J, Phillips TK, English M, McKnight J, Orrell C. Retention in a low-resource, high-burden cohort on antiretroviral therapy: Retrospective, longitudinal analysis comparing six measures of retention. *J Int AIDS Soc* 2025; **28:** e70046.

13 WHO. The role of HIV viral suppression in improving individual health and reducing transmission. World Health Organization, 2023.

14 Kleinbaum DG, Klein M. Extension of the Cox Proportional Hazards Model for time-dependent variables. In: Kleinbaum DG, Klein M, eds. Survival analysis: a self-learning text. Springer, 2012: 241–88.

15 Thompson CG, Kim RS, Aloe AM, Becker BJ. Extracting the variance inflation factor and other multicollinearity diagnostics from typical regression results. *Basic Appl Soc Psych* 2017; **39:** 81–90.

16 Chowdhury MZI, Turin TC. Variable selection strategies and its importance in clinical prediction modelling. *Fam Med Community Health* 2020; **8:** e000262.

17 Brian J, Waiswa S, Balinaine J, et al. Sustained retention, viral load suppression and their determinants among clients on HAART enrolled under differentiated service delivery models in eastern Uganda. *Res Sq* 2023; published online Oct 4. https://doi.org/10.21203/rs.3.rs-3377046/v1 (preprint).

18 Ibiloye O, Jwanle P, Masquillier C, et al. Long-term retention and predictors of attrition for key populations receiving antiretroviral treatment through community-based ART in Benue State Nigeria: a retrospective cohort study. *PLoS One* 2021; **16:** e0260557.

19 Rosen S, Grimsrud A, Ehrenkranz P, Katz I. Models of service delivery for optimizing a patient's first six months on antiretroviral therapy for HIV: an applied research agenda. *Gates Open Res* 2020; **4:** 116.

20 Teklu AM, Yirdaw KD. Patients who restart antiretroviral medication after interruption remain at high risk of unfavorable outcomes in Ethiopia. *BMC Health Serv Res* 2017; **17:** 247.

21 Ehrenkranz P, Rosen S, Boulle A, et al. The revolving door of HIV care: revising the service delivery cascade to achieve the UNAIDS 95–95–95 goals. *PLoS Med* 2021; **18:** e1003651.

22 Arendse KD, Walker C, Pfaff C, et al. Supporting re-engagement with HIV services after treatment interruption in South Africa: a mixed method program evaluation of MSF's Welcome Service. *Sci Rep* 2024; **14:** 7317.

23 Moolla H, Davies MA, Davies C, et al. The effect of care interruptions on mortality in adults resuming antiretroviral therapy. *AIDS* 2024; **38:** 1198–205.

24 Jiamsakul A, Kerr SJ, Ng OT, et al, and the TREAT Asia HIV Observational Database (TAHOD). Effects of unplanned treatment interruptions on HIV treatment failure—results from TAHOD. *Trop Med Int Health* 2016; **21:** 662–74.

25 Pry JM, Mwila C, Kapesa H, et al. Estimating potential silent transfer using baseline viral load measures among people presenting as new to HIV care in Lusaka, Zambia: a cross-sectional study. *BMJ Open* 2023; **13:** e070384.

26 Kanyike AM, Mwamba C, Keene C. Approaches to implementation of person-centred care for people living with HIV in low-resource settings. 2024. https://plus.iasociety.org/webcasts/approaches-implementation-person-centred-care-people-living-hiv-low-resource-settings (accessed June 16, 2024).

27 UNAIDS. The path that ends AIDS: UNAIDS global AIDS update 2023. 2023. https://www.unaids.org/en/resources/documents/2023/global-aids-update-2023 (accessed Dec 9, 2025).

28 Alhaj M, Amberbir A, Singogo E, et al. Retention on antiretroviral therapy during universal test and treat implementation in Zomba district, Malawi: a retrospective cohort study. *J Int AIDS Soc* 2019; **22:** e25239.

29 Phillips TK, Clouse K, Zerbe A, Orrell C, Abrams EJ, Myer L. Linkage to care, mobility and retention of HIV-positive postpartum women in antiretroviral therapy services in South Africa. *J Int AIDS Soc* 2018; **21** (suppl 4)**:** e25114.